Introduction to Software Engineering

Session #1

Importance of software

Software can have a huge impact in any aspect of society.

Where can you find software?

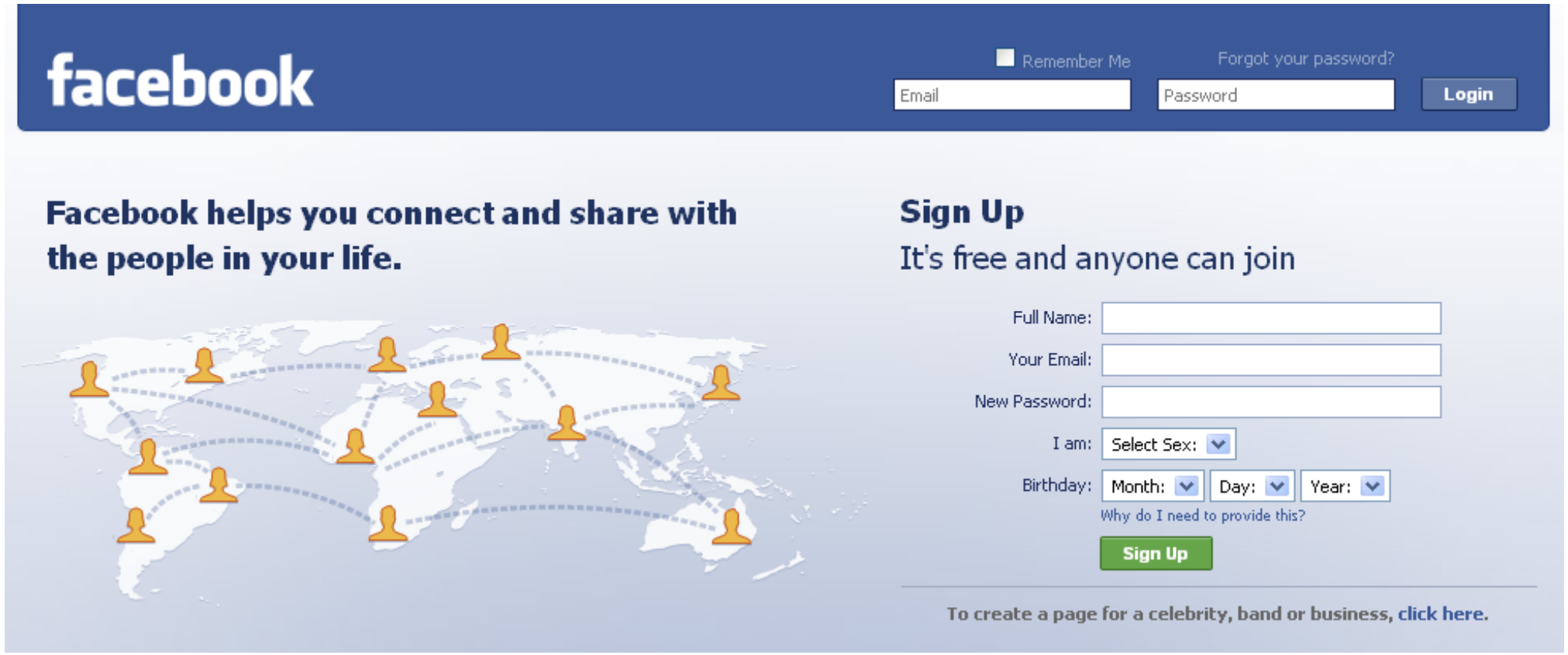




Some popular ones…



Some popular ones…



And even in…


Sony Ericsson
WALKMAN
UNOFFICIAL CLUB SONY ERICSSON


AT&T
9:42 AM
Text
Calendar
Photos
Camera
YouTube
Stocks
Maps
Weather
Clock
Calculator
Notes
Settings
iTunes
App Store
Phone
Mail
Safari
iPod

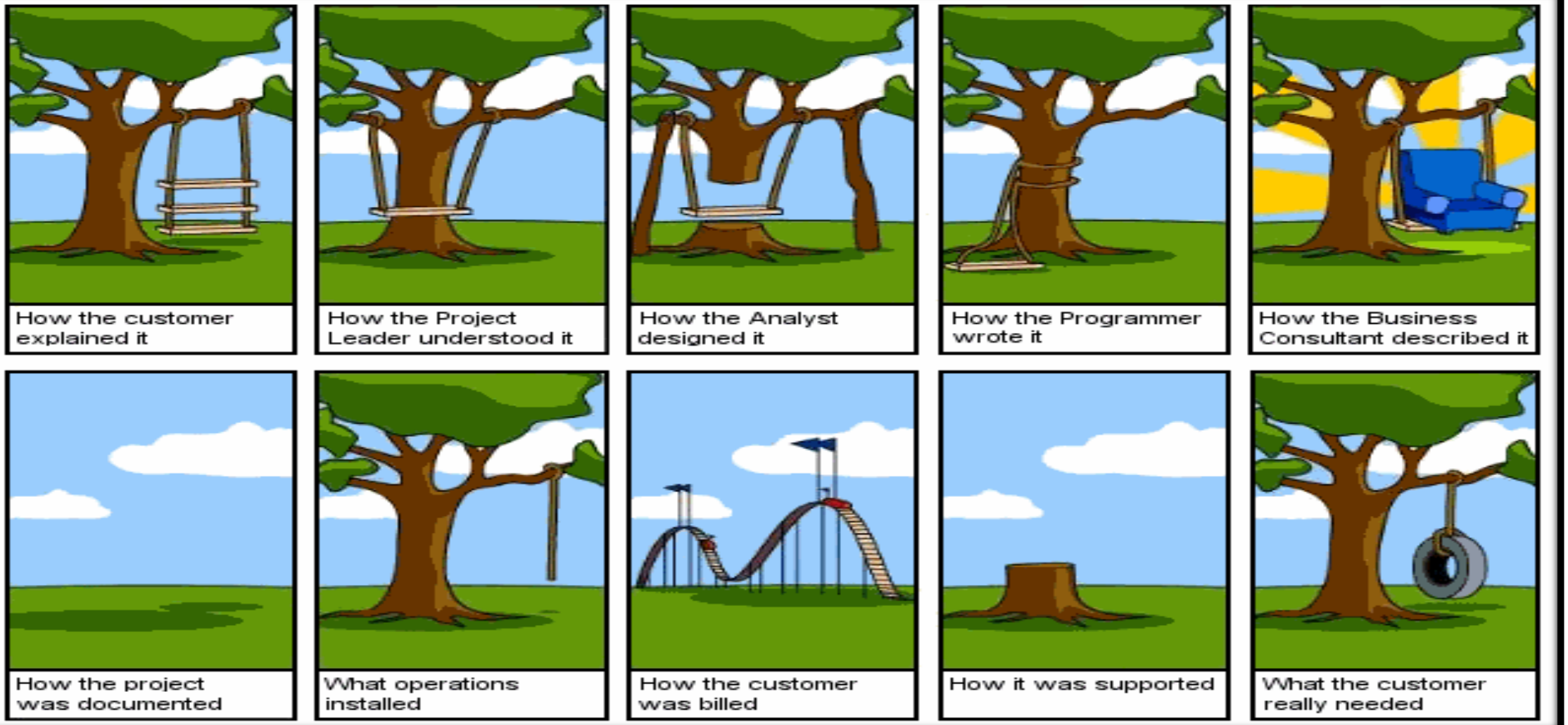
Conclusion

Software is Almost Everywhere.

Problems in software development

Common issues

- The final Software doesn´t fulfill the needs of the customer.
- Hard to extend and improve: if you want to add a functionality later is mission impossible.
- Bad documentation.
- Bad quality: frequent errors, hard to use, ...
- More time and costs than expected


How the customer explained it
How the Project Leader understood it
How the Analyst designed it
How the Programmer wrote it
How the Business Consultant described it
How the project was documented
What operations installed
How the customer was billed
How it was supported
What the customer really needed

Conclusion

Programming is NOT enough

- It is not enough to do your best: you must
- Know what to do, and THEN do your best.
- -- *W. Edwards Deming*

Solution

Software Engineering

- **What is it?**

The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software, and the study of these approaches; that is, the application of engineering to software.

Software Engineering

- **What is it?**

The study and application of methodologies to develop quality software that fulfill customer needs.

Software Engineering

Objetive:

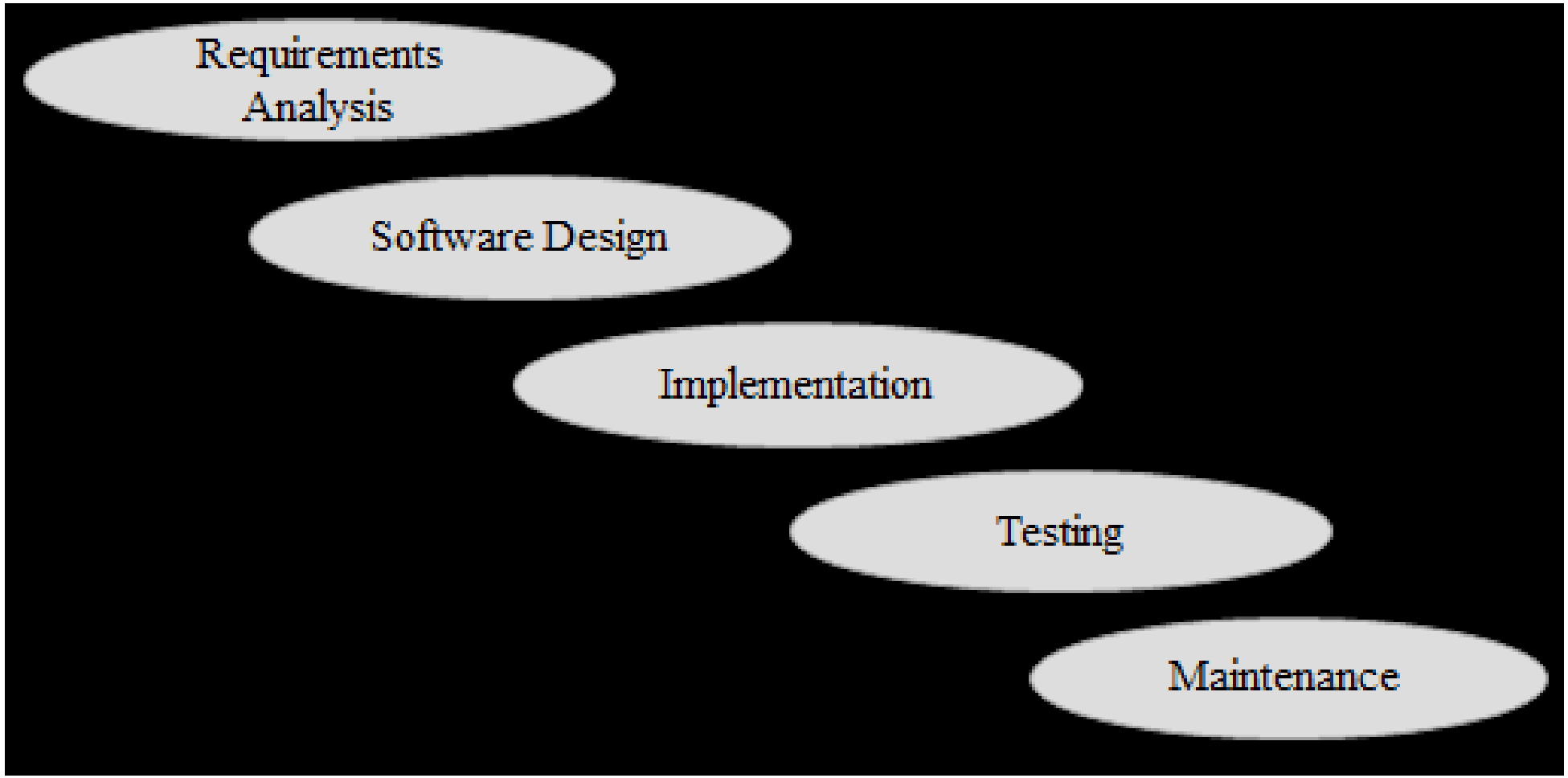
To produce software that is:

- On time: is deliver at the established date.

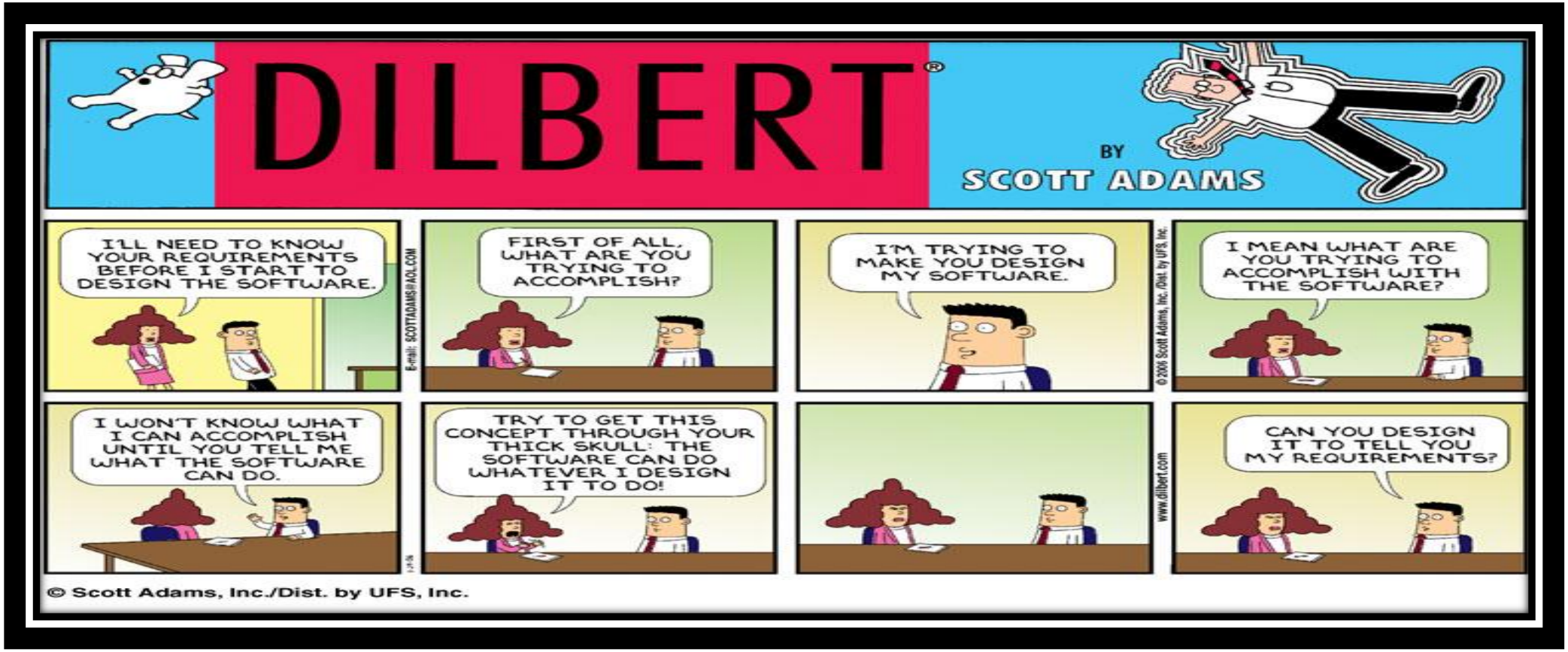
- Reliable: doesn´t crash.

- Complete: good documentation, fulfill customer needs.

Stages for software development

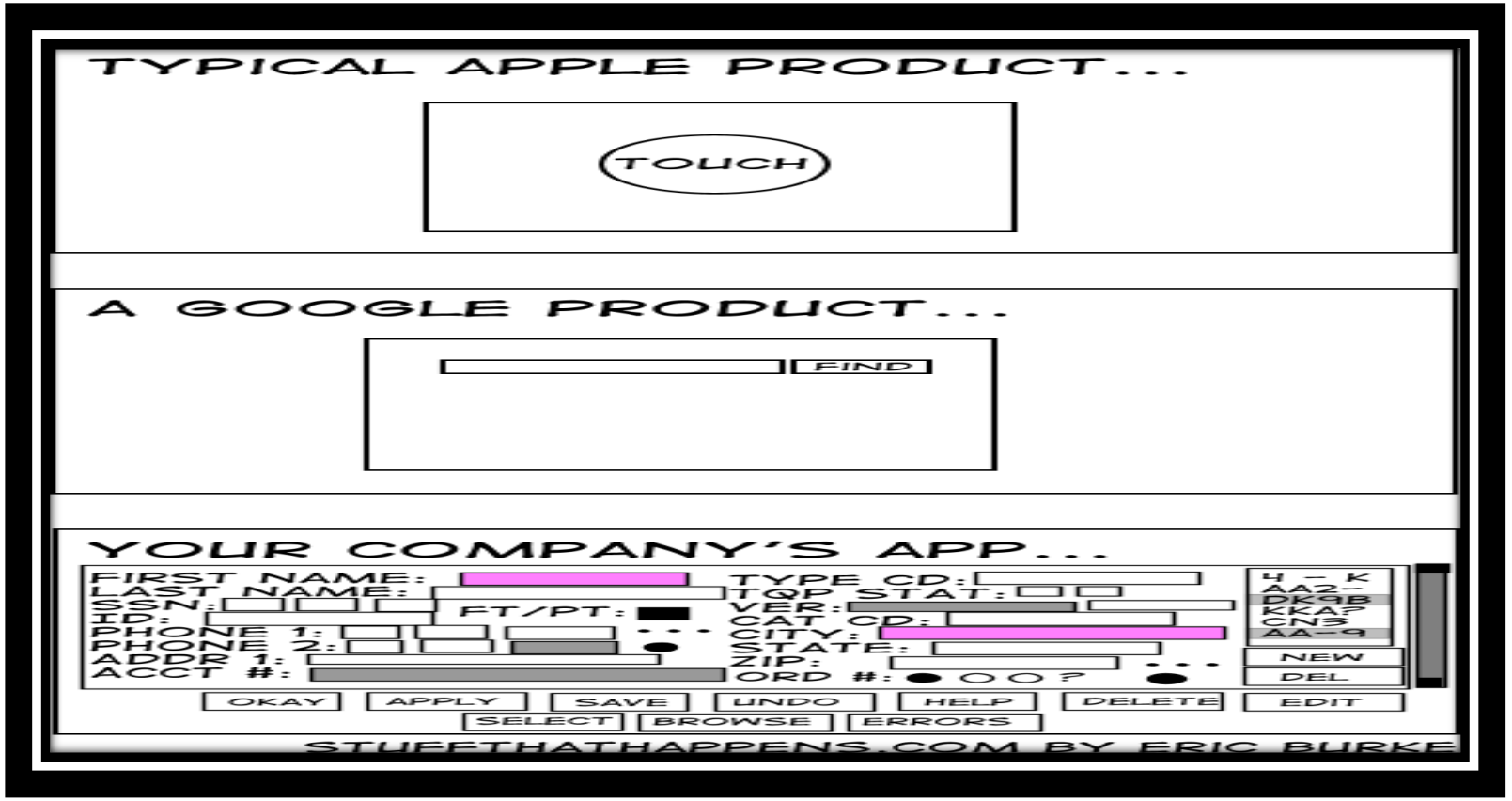


1. Requirements Analysis



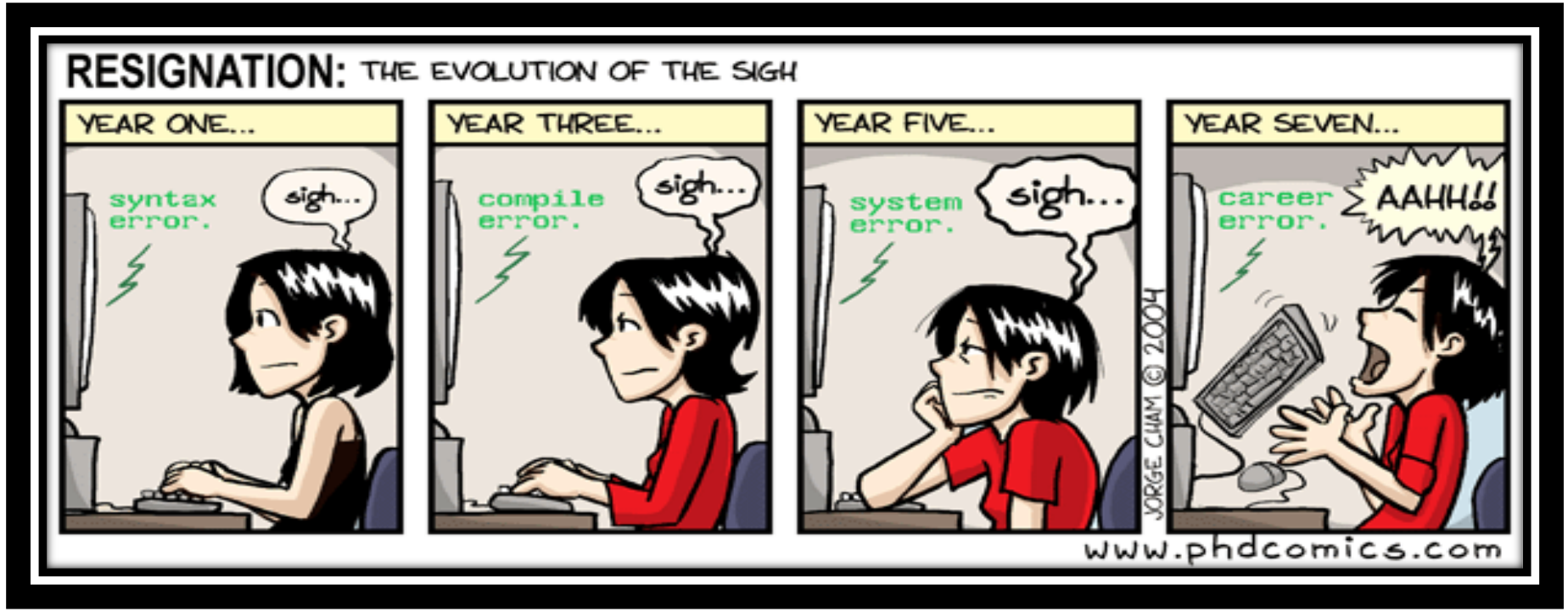
Find out what the client want the software to do

2. Design



Planning the software solution

Implementation



Code!!!

4. Testing



Executing the application trying to find software bugs

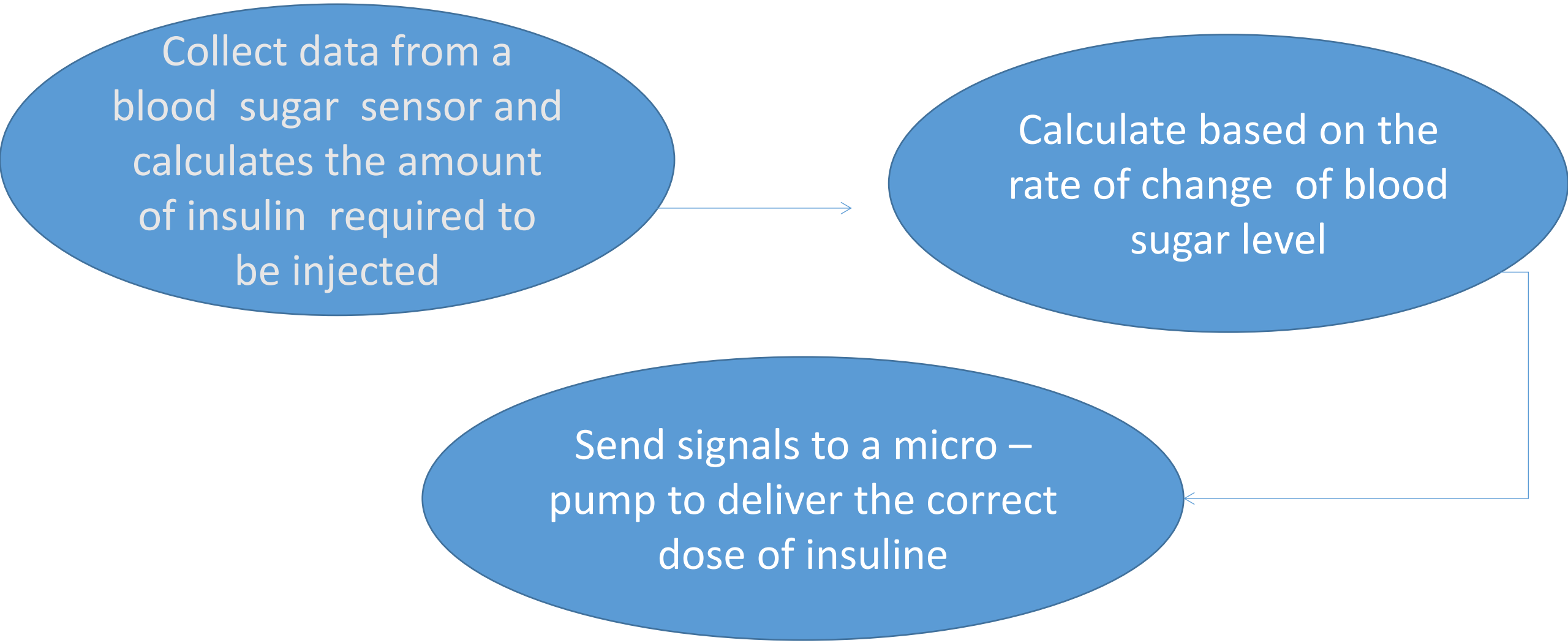
5. Maintenance

Any activity oriented to change an existing software product.

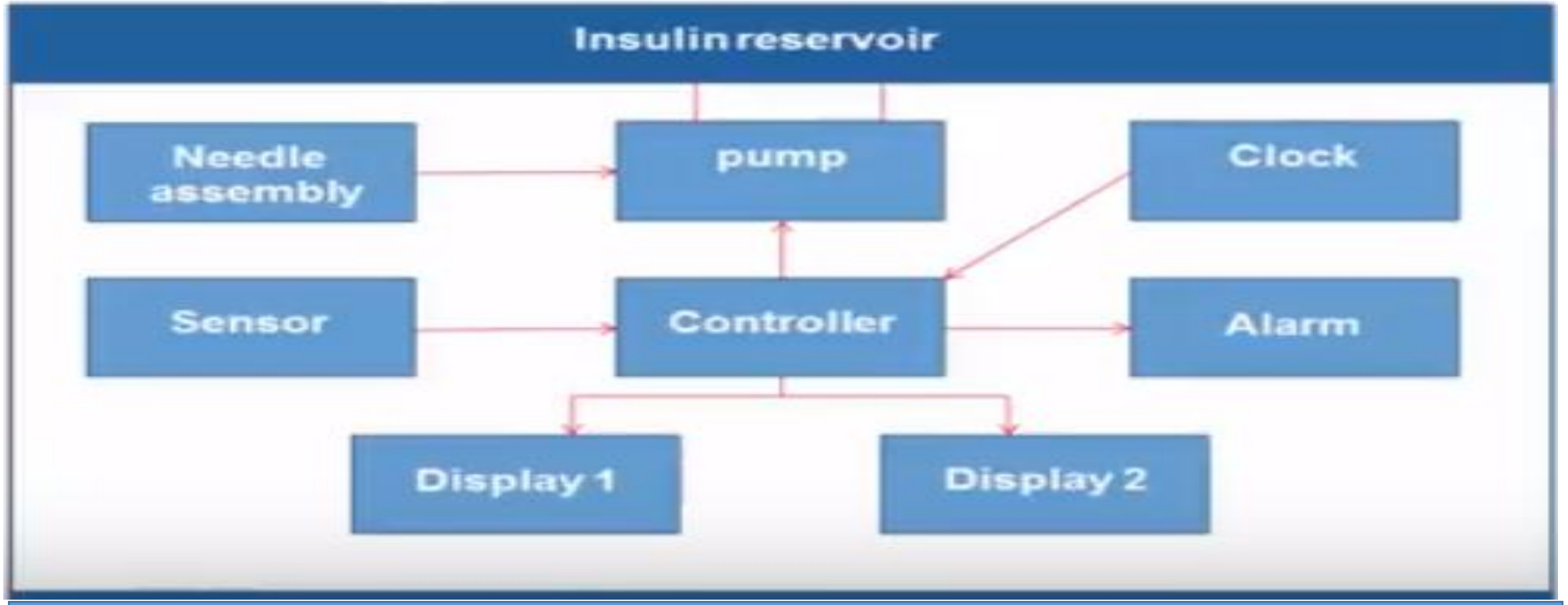
Case Study

Insulin pump control system

Insulin pump control system



Insulin pump hardware architecture



Thank you